3.2 Implicit Differentiation — teacher key

Question: How do you find the slope of a curve that is not a function?

Outline and pacing (42 min)

Tab	Priority	Minutes	If behind
start	1	2	never trimmed
warmup	1	4	never trimmed
predict	1	3	never trimmed
explore	3	8	skipped first
build	2	8	shrinks to 2 of 4
practice	1	8	never trimmed
check	1	4	never trimmed
Huddle h1	clock	minute 18	pulled in by the clock

Targets

- I can differentiate both sides of an equation with x and y.
- I can solve for dy/dx and find a slope at a point.
- I can write the tangent line at a point on the curve.

Predictions

- The circle $x^2 + y^2 = 25$ passes through $(3, 4)$ and $(3, -4)$. The tangent slopes at those two points are... **opposite in sign.** The upper arc falls at $x = 3$ and the lower arc rises there: the slopes are $-\frac{3}{4}$ and $+\frac{3}{4}$. One x cannot name the slope.
- Suppose $y = x^3$. Then $y^2 = x^6$, so $\dfrac{d}{dx}(y^2) = 6x^5$. Which expression gives $6x^5$? $2y \cdot \dfrac{dy}{dx}$. $2y \cdot \frac{dy}{dx} = 2x^3 \cdot 3x^2 = 6x^5$. The chain rule puts $\frac{dy}{dx}$ on every y-term.

Huddle script (answers in green)

Definitions

- **Implicit form:** An equation where x and y are **mixed** together.
- **The y-factor:** Every y-term gets a factor of **dy/dx**.
- **Solving:** Collect the dy/dx terms, **factor** it out, then divide.

Worked example

Find dy/dx for $x^2 + y^2 = 25$.

- Differentiate both sides: $2x + 2y \cdot dy/dx =$ **0**
- Isolate: $2y \cdot dy/dx =$ **−2x**
- Divide by 2y: $dy/dx =$ **−x/y**

Label the tangent

1. The point (3, 4) lies on the **curve**.
2. The **tangent line** touches the curve at that point.
3. Its slope at (3, 4) is **−3/4**.

Every level, with answers

Find dy/dx

Level 1. Differentiate both sides, then solve for $\frac{dy}{dx}$.
Answer: $\frac{4}{2y}$ $2y\,\dfrac{dy}{dx} = 4$, so $\dfrac{dy}{dx} = \dfrac{4}{2y}$.

Level 2. Differentiate both sides, then solve for $\frac{dy}{dx}$.
Answer: $\frac{-x}{2y}$ $2x + 4y\,\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = \dfrac{-2x}{4y} = -\dfrac{x}{2y}$.

Level 3. Differentiate both sides. Collect the $\frac{dy}{dx}$ terms, factor, and divide.
Answer: $\frac{5x^4-4x}{9y^2-1}$ $9y^2y' + 4x - y' = 5x^4$, so $y'(9y^2 - 1) = 5x^4 - 4x$.

Level 4. Differentiate both sides. The right side needs the product rule.
Answer: $\frac{6y-3x^2}{3y^2-6x}$ $3x^2 + 3y^2y' = 6y + 6x\,y'$, so $y'(3y^2 - 6x) = 6y - 3x^2$.

Differentiate a y-term

Level 1. $\dfrac{d}{dx}\left(y^3\right)$ equals...
(A) 3y² · dy/dx (B) 3x² (C) 3y² · dx/dy (D) 3y²
Answer: A Power rule gives $3y^2$. The chain rule adds the factor $\frac{dy}{dx}$.

Level 2. $\dfrac{d}{dx}\left(6y^2\right)$ equals...
(A) 12y (B) 6y · dy/dx **(C) 12y · dy/dx** (D) 12x
Answer: C Coefficient: 6 times 2 is 12. Then the factor $\frac{dy}{dx}$.

Level 3. $\dfrac{d}{dx}\left(x^3y\right)$ equals...
(A) 3x²y (B) 3x² · dy/dx (C) x³ · dy/dx **(D) 3x²y + x³ · dy/dx**
Answer: D Product rule: (derivative of the first)(second) + (first)(derivative of the second). The y part brings $\frac{dy}{dx}$.

Level 4. $\dfrac{d}{dx}\left(x^2e^y\right)$ equals...
(A) x² · e^y · dy/dx (B) 2x · e^y **(C) 2x · e^y + x² · e^y · dy/dx** (D) 2x · e^y · dy/dx
Answer: C Product rule: $2x\,e^y$ plus $x^2e^y\frac{dy}{dx}$.

Slope at a point

Level 1. The point $(6, -8)$ is on the curve $x^2 + y^2 = 100$. Find $\dfrac{dy}{dx}$ there.
Answer: 0.75 Implicit differentiation gives a slope formula in x and y. At $(6, -8)$ it equals $\dfrac{3}{4}$.

Level 2. The point $(-1, 2)$ is on the curve $x^2 + 2y^2 = 9$. Find $\dfrac{dy}{dx}$ there.

Answer: 0.25 Implicit differentiation gives a slope formula in x and y. At $(-1, 2)$ it equals $\dfrac{1}{4}$.

Level 3. The point $(4, -1)$ is on the curve $x^2 + xy + y^2 = 13$. Find $\dfrac{dy}{dx}$ there.

Answer: -3.5 Implicit differentiation gives a slope formula in x and y. At $(4, -1)$ it equals $\dfrac{-7}{2}$.

Level 4. The point $(1, 4)$ is on the curve $6xy = 8 + y^2$. Find $\dfrac{dy}{dx}$ there.

Answer: 12 Implicit differentiation gives a slope formula in x and y. At $(1, 4)$ it equals 12.

Tangent line

Level 1. Write the tangent line to the curve at the point.

Answer: $\frac{5}{12}(x - (-5)) + 12$ Slope at the point: $\dfrac{5}{12}$. Then $y = m(x - (-5)) + (12)$.

Level 2. Write the tangent line to the curve at the point.

Answer: $\frac{1}{4}(x - (-1)) + 2$ Slope at the point: $\dfrac{1}{4}$. Then $y = m(x - (-1)) + (2)$.

Level 3. Write the tangent line to the curve at the point.

Answer: $\frac{-7}{2}(x - 4) - 1$ Slope at the point: $\dfrac{-7}{2}$. Then $y = m(x - (4)) + (-1)$.

Level 4. Write the tangent line to the curve at the point. Keep the radicals exact.

Answer: $\frac{3}{\sqrt{7}}(x - 3) - \sqrt{7}$ Slope: $-\dfrac{3}{-\sqrt{7}}$. Then $y = m(x - 3) + (-\sqrt{7})$.

Common misconceptions

Students say	Say instead
d/dx (y^2) = 2y.	y depends on x. Multiply by dy/dx: 2y · dy/dx.
d/dx (xy) = y.	It is a product: y + x · dy/dx.
Differentiate only the left side.	Both sides. A constant on the right becomes 0.
The slope is –x/y, so plug in x only.	The slope needs both coordinates of the point on the curve.